



University of Rwanda
College of Sciences and Technology
School of Science
Department of Mathematics

Final Exam - Academic year 2024-2025
MAT1166: Mathematics For Engineers I
First Semester

Instructions

1. Complete the cover page clearly and follow carefully the instructions below.
2. Fill the exam attendance slip at the back of the booklet and tear it for your own record.
3. Don't forget that cell phones are not allowed inside the examination room.
4. This paper has SECTION A and SECTION B.
5. Questions in SECTION A are compulsory.
6. Answer any TWO questions from SECTION B.
7. No written materials allowed in examination room.
8. Write all your answers in the answer booklet provided.
9. Don't forget to write your registration number.

Section A (Compulsory, 20 marks)

Question 1 (4 marks)

- (a) Given an arithmetic progression $\{2, 5, 8, 11, \dots\}$. Find a general formula for u_n and then find the sum of the first 20 terms of the sequence. (2 marks) $S = 610$
- (b) Determine whether the following function $f(x)$ is continuous at $x = 0$ or not. Justify your answer. (2 marks)

$$f(x) = \begin{cases} \frac{\sin(2x)}{x}, & x \neq 0, \\ 2, & x = 0. \end{cases}$$

Question 2 (4 marks)

- (a) Find the derivative of $f(x) = x^3 - 2x^2 + x - 10$. (1 marks)
- (b) Determine the critical points of $f(x)$ and classify each as a local maximum, local minimum, or neither. (3 marks)

$$\frac{1}{n} (u_1 + u_n)$$

1

$$\frac{2+19}{2}$$

$$60 - 12 = 48$$

2

Question 3 (4 marks)

- (a) A quantity surveyor encounters the improper integral

$$\int_0^{\infty} x e^{-x^2} dx$$

while modeling resource decay over time. Express this integral as a limit and evaluate its value. State whether the integral converges or diverges. (2 marks)

- (b) Find $\frac{dy}{dx}$ if $y \sin x = x^3 + \cos y$. (2 marks)

Question 4 (4 marks)

- (a) Find the value of x so that the following matrix is singular. (2 marks)

$$A = \begin{pmatrix} 1 & 0 & -x \\ -1 & 3 & 1 \\ 0 & 2x & -4 \end{pmatrix}.$$

- (b) Evaluate the following limit (2 marks)

$$\lim_{x \rightarrow \infty} \left(1 - \frac{3}{x}\right)^{-x}$$

Question 5 (4 marks)

Given two vectors in three-dimensional space, $\vec{u} = 2\vec{i} - \vec{j} + 3\vec{k}$ and $\vec{v} = \vec{i} + 4\vec{j}$.

- (a) Compute the scalar (dot) product $\vec{u} \cdot \vec{v}$ and the vector (cross) product $\vec{u} \times \vec{v}$. (3 marks)
- (b) Use the result of the cross product found in sub-question 5.(a) to find the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$. (1 mark)

Section B (Choose Only Two, 2×15 marks = 30 marks)

Question 6 (15 marks)

- (a) Consider the integral

$$\int f(x) dx,$$

where $f(x) = \frac{x+5}{x^2+2x+8}$.

This integral appears in an engineering setting where you need to find the work done by a variable force.

- (i) By using a suitable integration technique, evaluate the integral. (7 marks)
- (ii) Suppose that the integral in sub-question (i) represents the work done by a force $F(x)$ moving an object from $x = 0$ to $x = 3$. Deduce the value of $F(x)$ by evaluating the definite integral (3 marks)

$$\int_0^3 f(x) dx.$$

- (b) Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$. (5 marks)

Question 7 (15 marks)

- (a) Using an appropriate test of convergence, determine whether the following series converges or diverges (5 marks)

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}.$$

(b) Power series and Maclaurin and Taylor expansions are widely used in computing approximate solutions to engineering problems.

(i) Determine the radius and interval of convergence of the power series (5 marks)

$$\sum_{n=0}^{\infty} \frac{(x-3)^n}{4^n}$$

(ii) Find the Maclaurin polynomial $P_4(x)$ for $f(x) = e^{-2x}$. (5 marks)

Question 8 (15 marks)

An engineer is analyzing the energy consumption of a smart factory that uses three types of machines: A, B, and C. The energy consumption rate (units per hour) of each machine is unknown. However, the total energy consumption for several scenarios is recorded as follows:

Scenario 1: 5 hours of Machine A, 3 hours of Machine B, and 4 hours of Machine C consumed 520 energy units. $x=35$

Scenario 2: 6 hours of Machine A, 2 hours of Machine B, and 3 hours of Machine C consumed 450 energy units. $y=76$

Scenario 3: 4 hours of Machine A, 5 hours of Machine B, and 2 hours of Machine C consumed 580 energy units. $z=29$

The objective is to determine the energy consumption rate (units per hour) of each machine.

- Translate the problem into a system of linear equations by defining the energy consumption rates of the machines as x , y , and z (units per hour) of the machine A, B, and C, respectively, and write the system in matrix form $AX = B$. (3 marks)
- Use any method of your choice to solve the system found in sub-question (a) for x , y , z and conclude. Please round the results to the nearest whole number. (7 marks)
- Verify that the set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ is an orthogonal set in $\mathbb{R}^3$? (2.5 marks)
 - In view of i) find the corresponding orthonormal set in $\mathbb{R}^3$ (2.5 marks).

Question 9 (15 marks)

(a) If $\vec{a} = \langle 2, 1, -1 \rangle$, $\vec{b} = \langle 3, 0, 5 \rangle$ and $\vec{c} = \langle -3, 4, 1 \rangle$, compute the following (5 marks)

$$\vec{c} \cdot (\vec{a} \times \vec{b})$$

(b) Eigenvalues and eigenvectors are essential tools for analyzing vibration modes in structures and assessing stability in mechanical and electrical systems. The sign of an eigenvalue provides insight into system behavior: **Positive eigenvalues** ($\lambda > 0$) indicate stable behavior. In vibration analysis, they correspond to modes where the structure vibrates safely, without risk of catastrophic failure. **Zero eigenvalues** ($\lambda = 0$) signify a system that is neutrally stable, neither fully stable nor unstable. **Negative eigenvalues** ($\lambda < 0$) indicate instability, where vibrations can grow uncontrollably, potentially causing structural failure. The eigenvectors associated with these eigenvalues represent the mode shapes of vibration. So, for a structure to be considered safe, all eigenvalues must be positive.

In a critical buckling load analysis, consider the stiffness matrix of a column:

$$M = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$$

$$\lambda_1 = 5 \quad \lambda_2 = 2 \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- Determine the critical buckling load (eigenvalues) of the matrix M , and evaluate the safety of the structural system based on the signs of the calculated eigenvalues. (5 marks)
- Find the mode shapes (eigenvectors) corresponding to each critical buckling load (eigenvalue). (5 marks)